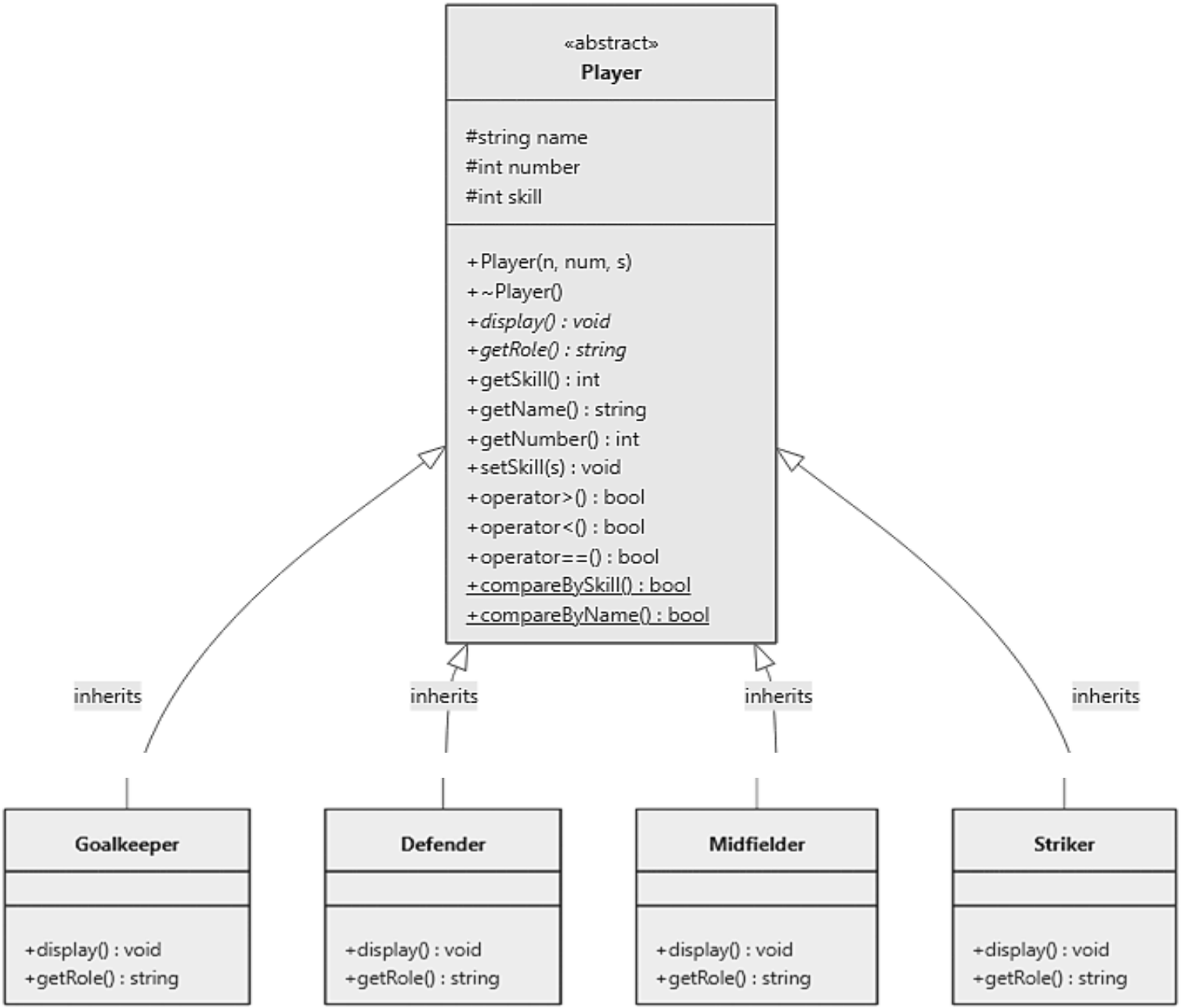
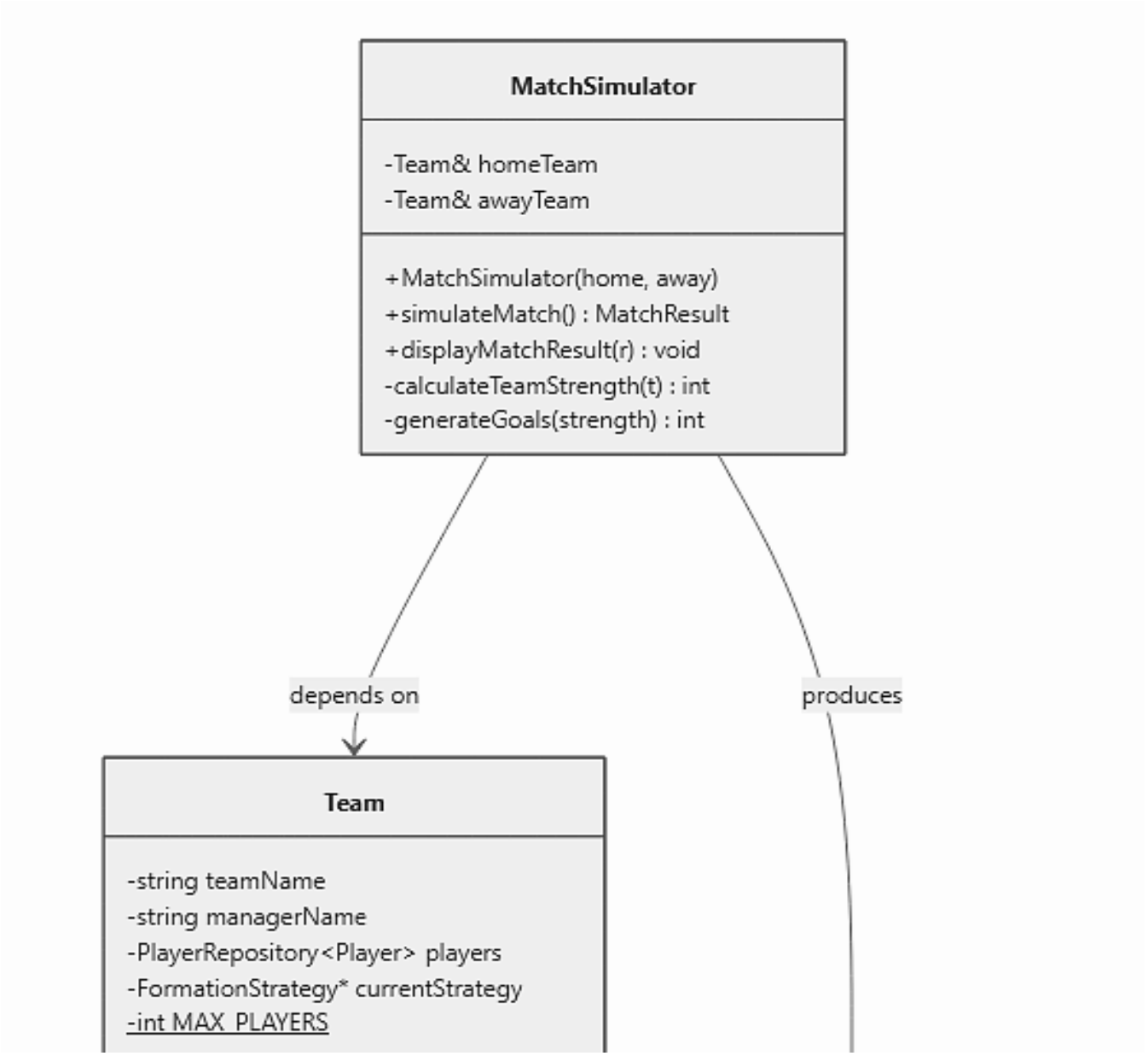
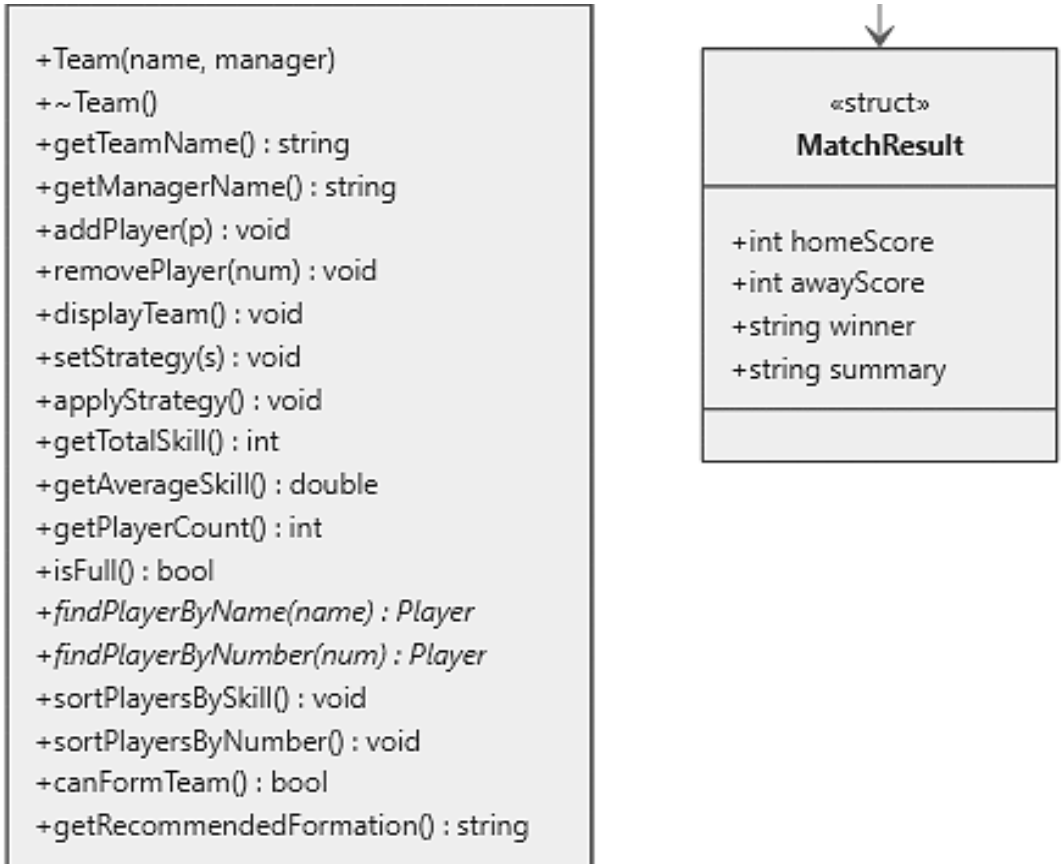
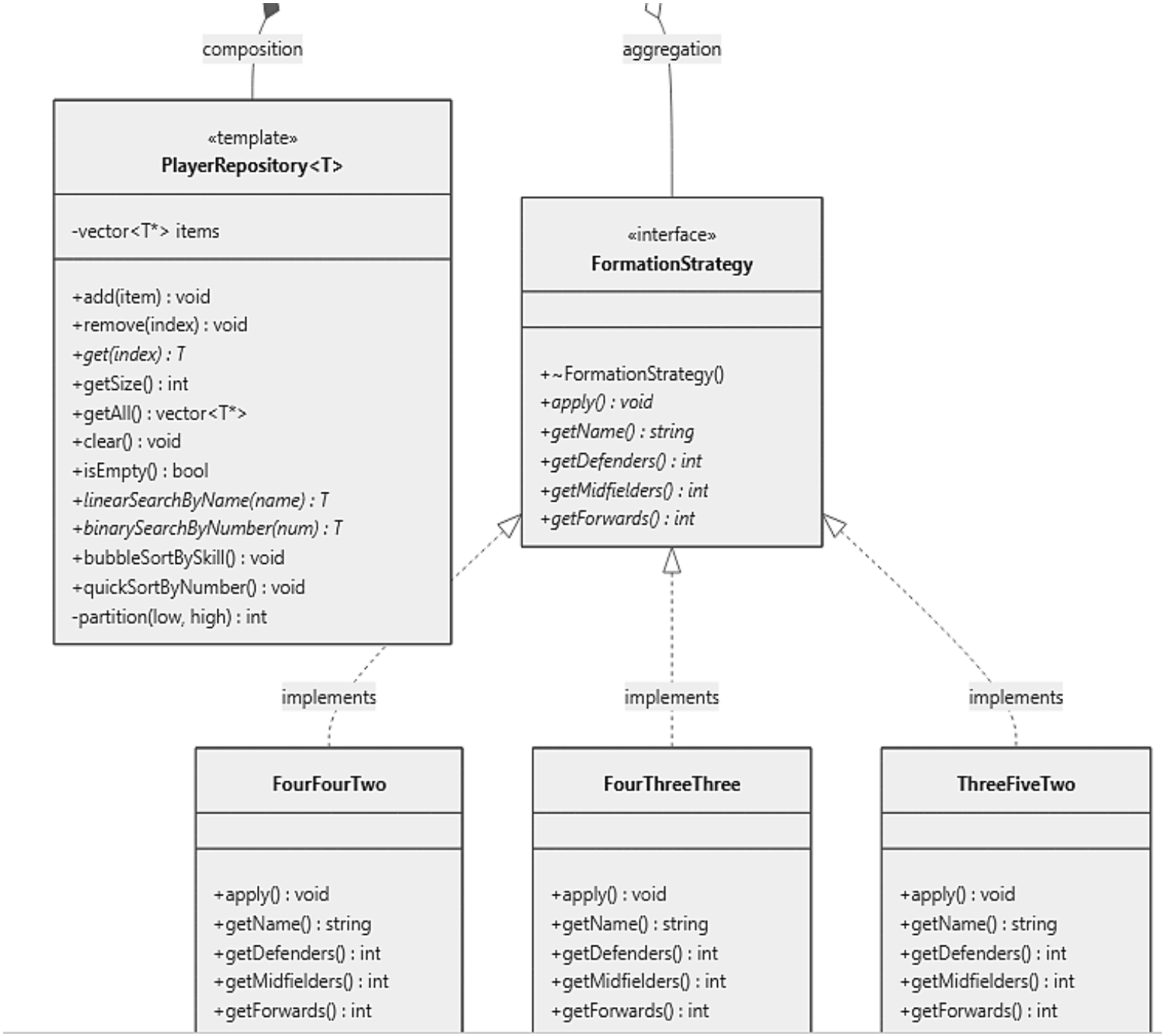

«abstract»
Player
#string name
#int number
#int skill
+Player(n, num, s)
+~Player()
+display() : void
+getRole() : string
+getSkill() : int
+getName() : string
+getNumber() : int
+setSkill(s) : void
+operator>() : bool
+operator<() : bool
+operator==() : bool
+compareBySkill() : bool
+compareByName() : bool
inherits
inherits
inherits
inherits
Goalkeeper
+display() : void
+getRole() : string
Defender
+display() : void
+getRole() : string
Midfielder
+display() : void
+getRole() : string
Striker
+display() : void
+getRole() : string


MatchSimulator
-Team& homeTeam
-Team& awayTeam
+MatchSimulator(home, away)
+simulateMatch() : MatchResult
+displayMatchResult(r) : void
-calculateTeamStrength(t) : int
-generateGoals(strength) : int
depends on
produces
Team
-string teamName
-string managerName
-PlayerRepository<Player> players
-FormationStrategy* currentStrategy
-int MAX_PLAYERS


+Team(name, manager)
+~Team()
+getTeamName() : string
+getManagerName() : string
+addPlayer(p) : void
+removePlayer(num) : void
+displayTeam() : void
+setStrategy(s) : void
+applyStrategy() : void
+getTotalSkill() : int
+getAverageSkill() : double
+getPlayerCount() : int
+isFull() : bool
+findPlayerByName(name) : Player
+findPlayerByNumber(num) : Player
+sortPlayersBySkill() : void
+sortPlayersByNumber() : void
+canFormTeam() : bool
+getRecommendedFormation() : string
«struct»
MatchResult
+int homeScore
+int awayScore
+string winner
+string summary


composition
aggregation
«template»
PlayerRepository<T>
-vector<T*> items
+add(item) : void
+remove(index) : void
+get(index) : T
+getSize() : int
+getAll() : vector<T*>
+clear() : void
+isEmpty() : bool
+linearSearchByName(name) : T
+binarySearchByNumber(num) : T
+bubbleSortBySkill() : void
+quickSortByNumber() : void
-partition(low, high) : int
«interface»
FormationStrategy
+~FormationStrategy()
+apply() : void
+getName() : string
+getDefenders() : int
+getMidfielders() : int
+getForwards() : int
implements
implements
implements
FourFourTwo
+apply() : void
+getName() : string
+getDefenders() : int
+getMidfielders() : int
+getForwards() : int
FourThreeThree
+apply() : void
+getName() : string
+getDefenders() : int
+getMidfielders() : int
+getForwards() : int
ThreeFiveTwo
+apply() : void
+getName() : string
+getDefenders() : int
+getMidfielders() : int
+getForwards() : int